

Top-Down Pooling and Anticipatory Exposure

Abstract

The standard signaling story explains separation from below: low types cannot afford to imitate high types. This paper identifies the mirror failure mode in a two-type dynamic psychological signaling model. Sending an ambitious signal increases current persuasive value, but it also creates a one-period reputational exposure: today's action can be repriced tomorrow if the judgment state turns to high scrutiny.

Each type has a primitive self-model accuracy level, and a higher true type can carry higher exposure risk under adverse repricing. The high-scrutiny transition probability is $q = \Pr(j_{t+1} = H \mid j_t = L)$. The current rent from an ambitious signal for type θ is

$$D_\theta^L(q) = B_\theta - \beta q \mu_H \Phi_\theta.$$

The key is that $\Phi_H > \Phi_L$ and $B_H > 0$ can coexist.

This creates top-down collapse. Let

$$q_{IC} = \frac{B_H}{\beta \mu_H \Phi_H}, \quad q_{D1} = \text{the D1 willingness threshold}, \quad q^* = \max\{q_{IC}, q_{D1}\}.$$

For $q > q^*$, a binary-signal pure strategy that separates is not self-confirming: the high type would withdraw from s_A even on-path, and D1 then removes the high-type interpretation of off-path s_A .

The paper first proves a D1-closed fixed-point selection result in the binary class: the unique D1-consistent assessment is pooling on the safe signal, while the ambitious signal becomes a signaling dead-end from above. A continuous-type extension in Section 5 describes the observable shadow as support polarization of ambitious senders, not simple mean reversion. A microfounded dynamic section then shows this collapse is robust to trust-capital and reinforcement effects: they shift margins, but bounded endogenous drift cannot generate automatic restoration. A continuous-time population counterpart in Section 7 translates the same D1-closed exit map into top-end depletion of the ambitious stock.

1 Introduction

In standard costly signaling, separation is sustained from below: low types cannot afford high-cost signaling, so they do not mimic high types. The model here studies the opposite failure mode. A good type may still benefit from an ambitious signal today, but can rationally retreat because that action creates a one-period exposure that can be repriced against her future credibility.

The mechanism is dynamic and belief-driven. The ambitious action has two effects. It is informative in the current period, but it also claims something about the sender's self-model. If high scrutiny arrives tomorrow, the claim can be reinterpreted as evidence of weaker self-understanding and thus impose an additional penalty. This is adverse repricing, not a true-type audit.

The primitives are minimal: two types, two sender signals, and an exogenous binary judgment state. The sender observes the low-scrutiny risk $q = \Pr(j_{t+1} = H \mid j_t = L)$ before acting. If she sends the ambitious signal, tomorrow's state can impose an extra epistemic cost. If she sends the pooling-safe action, that one-period shadow disappears.

The model has four moving parts.

1. A D1-closed fixed-point core that shows a top-down reversal: high types exit separation first when q is large, and D1 then removes the high-type meaning of off-path ambition.
2. An observable-implications section that translates the binary collapse into support polarization in a richer type space.
3. A dynamic forward-invariance result that adds trust-capital/semantic dynamics and shows endogenous accumulation does not automatically repair the collapse.
4. A continuous-time population counterpart that restates top-end nonexistence as loss of a positive stationary occupancy profile.

The first stage identifies how adverse repricing plus a D1-style off-path map can select pooling from above. It closes the belief loop in binary form: a self-referential interpretation where high occupancy of the ambitious signal implies high-type meaning, but then current IC may already fail, and off-path reinterpretation then collapses that meaning. The second stage records the observable cross-sectional shadow. The third stage asks whether time and dynamics can reverse this outcome. The final stage gives the distribution-level counterpart.

The result is defensive: no new stopping phenomenon appears. D1-closed collapse does not self-correct through learning or accumulation alone.

Section 2 introduces primitives and the one-period IC comparison. Section 3 derives the exposure ranking from primitive self-accuracy levels. Section 4 proves the D1-closed fixed-point theorem and the threshold $q^* = \max\{q_{IC}, q_{D1}\}$. Section 5 presents the observable polarization implication. Section 6 gives the dynamic forward-invariance result. Section 7 gives the continuous-time population counterpart.

2 Primal Structure and Benchmark

This paper studies a signaling environment with two sender types and two signals, where the same signal that can separate today also creates a near-term reputation risk that may be repriced tomorrow. The baseline is the dynamic extension of Spence logic needed for the fixed-point reversal.

2.1 Types, signals, and judgment states

There is a sender with type

$$\theta \in \{\theta_L, \theta_H\}, \quad \theta_H > \theta_L.$$

The receiver does not observe the type.

The sender chooses one of two signals

$$s \in \{s_A, s_P\}.$$

Signal s_A is the ambitious action, and s_P is the pooling-safe action.

Judgment intensity follows a public Markov state

$$j_t \in \{L, H\}.$$

State L is ordinary evaluation, and state H is high-scrutiny evaluation. Let

$$\mu_H > \mu_L > 0$$

denote the scrutiny multipliers and

$$q := \Pr(j_{t+1} = H \mid j_t = L)$$

be the transition probability from low to high. The sender observes q before choosing a signal.

2.2 One-period exposure shadow

Choosing s_A creates a one-period exposure state that can be repriced tomorrow. Let

$$X_t \in \{0, 1\}$$

indicate whether last period's action is exposed in the next period. If $s_t = s_A$, then $X_{t+1} = 1$; if $s_t = s_P$, then $X_{t+1} = 0$. In this baseline the exposure horizon is one period: choosing the safe signal today removes the next-period exposure created by a yesterday ambitious action.

If $X_{t+1} = 1$ and $j_{t+1} = H$, the sender incurs a future epistemic penalty proportional to a true-type exposure index. This is the anticipatory shadow that drives behavior in the low state.

2.3 Current payoff and the anticipatory rent

Let B_θ denote the current net separation surplus from s_A versus s_P in state L when the receiver interprets the ambitious signal in the high-type way. The low-state payoff gain from sending s_A is

$$D_\theta^L(q) = B_\theta - \beta q \mu_H \Phi_\theta,$$

where $\Phi_\theta > 0$ is the true-type exposure cost derived from primitive self-model accuracy (Section 3), and $\beta \in (0, 1)$ is the discount factor.

The type- θ IC threshold is

$$q_\theta^* := \frac{B_\theta}{\beta \mu_H \Phi_\theta},$$

whenever $B_\theta > 0$. Type θ is willing to send s_A in the low-scrutiny state iff

$$q < q_\theta^*.$$

2.4 Static benchmark

When $q = 0$, current risk from future repricing is absent. The model then collapses to a one-shot separating benchmark.

Assumption 1 (Static separation).

$$B_H > 0, \quad B_L < 0.$$

Under Assumption 1, with no anticipated repricing s_A is uniquely attractive only to the high type.

2.5 Anticipatory reversal object

The model's distinctive case is high-type exposure vulnerability: the same primitive that makes ambitious separation valuable can make the high type more exposed to adverse repricing. Assumption 1 therefore identifies the low-scrutiny rent before future exposure is priced; it does not guarantee that the high type keeps using s_A when the transition probability into high scrutiny is large. Section 4 defines the IC and D1 thresholds and shows how their maximum selects safe pooling.

3 Psychological Exposure and Bayesian Surprise

The benchmark environment in Section 2 treats the ambitious signal s_A as a costly signal of ability. This section adds the psychological object that distinguishes the model from standard costly signaling. The ambitious signal does not only reveal information about the sender's type. It also exposes the sender's self-model. A sender who chooses s_A represents herself as someone whose private assessment of her own capacity is accurate enough to justify taking the ambitious action. The social cost of later reinterpretation is therefore not only a material or reputational loss. It is also an epistemic loss: observers revise their beliefs about whether the sender correctly understood her own type.

This is a Bayesian mechanism with a second-order signal: observers infer type from the action, and infer whether the sender accurately understood the implications of being observed from the act of strategic deployment.

The section makes three clarifications. First, the exposure index Φ_θ is a true-type-indexed primitive object, derived from primitive self-model-accuracy levels. It is not an equilibrium object generated by a previous separating history. Second, the relevant channel is adverse repricing, not truth verification. A high-precision audit would tend to confirm genuinely accurate high types; that is not the technology studied here. Third, the D1 argument later uses Φ_θ as a cost attached to the sender's true type, not as a cost attached to the receiver's off-path attribution. This keeps the exposure technology separate from the belief refinement that interprets deviations. The section also keeps the mechanism non-political: all channels are Bayesian updates and reputation geometry, without political-moral interpretation.

3.1 Belief-dependent exposure

Let $j \in \{L, H\}$ denote the current judgment-intensity state, with associated scrutiny multiplier $\mu_j > 0$. The low state L represents ordinary evaluation and the high state H represents intensified evaluation. The state variable j is not a political primitive. It is an exogenous state of the social evaluation technology.

After s_A , observers form beliefs not only about the sender's ability θ , but also about the accuracy of the sender's self-model. Write

$$m \in \{0, 1\}$$

for the event that the sender's self-model is accurate, where $m = 1$ denotes accurate self-assessment. The sender cares not only about what observers infer about θ , but also about what observers infer about the sender's own assessment of θ .

For a finite hierarchy of beliefs $(\mu^2, \dots, \mu^k)$, let

$$L(\mu^2, \dots, \mu^k; j, x)$$

denote the epistemic penalty in judgment state j and exposure state $x \in \{0, 1\}$. The exposure state records whether the sender previously chose the ambitious signal. Choosing s_A sets $x = 1$ next period; choosing s_P clears exposure next period.

The total one-period psychological payoff can be written as

$$u_\theta(s, j, x; \mu^2, \dots, \mu^k) = v_\theta(s) - c_\theta(s) - \mu_j L(\mu^2, \dots, \mu^k; j, x),$$

where $v_\theta(s)$ is the ordinary reputational or material payoff from the signal, $c_\theta(s)$ is the intrinsic cost of the signal, and the final term is the judgment-weighted epistemic penalty.

The next assumption is purely regularity. It rules out discontinuities inside a fixed judgment state. The discontinuities studied below will come from changes in the transition probability into high scrutiny, not from discontinuous psychological utility within a regime.

Assumption 2 (Regime-conditional continuity). *For each fixed $j \in \{L, H\}$ and $x \in \{0, 1\}$, the function*

$$L(\mu^2, \dots, \mu^k; j, x)$$

is continuous in the relevant finite hierarchy of higher-order beliefs. The action set, type space, and exposure state space are finite.

Lemma 1 (Regime-conditional continuity). *Under Assumption 2, for each fixed judgment-intensity state j , the induced psychological payoff is continuous in the relevant higher-order belief hierarchy. Hence the regime-conditional psychological game satisfies the standard continuity requirement for belief-dependent payoffs. On any open region where the relevant incentive constraints are strict, the associated pure-strategy equilibrium selection is locally constant.*

Proof. Fix j and x . The ordinary payoff terms $v_\theta(s)$ and $c_\theta(s)$ are defined on a finite action and type space. The multiplier μ_j is constant within the state j . By Assumption 2, the epistemic penalty $L(\mu^2, \dots, \mu^k; j, x)$ is continuous in the relevant belief hierarchy. Multiplication by the fixed scalar μ_j and addition of finite ordinary payoff terms preserve continuity. Thus, the regime-conditional psychological payoff is continuous.

Since the type and action spaces are finite, strict incentive inequalities are preserved under sufficiently small perturbations of beliefs and payoff parameters within a fixed state. Therefore, wherever the relevant incentive constraints are strict, the same pure-strategy profile remains selected in a neighborhood of the primitive. Any discontinuous change in equilibrium behavior must therefore come from a change in the relevant state, transition probability, or binding incentive constraint, rather than from a discontinuity of belief-dependent utility within the fixed state. $\square$

3.2 Primitive self-accuracy levels

The exposure index is built from a level channel, not a precision channel. Let

$$p_\theta := \Pr(m = 1 \mid \theta)$$

denote the primitive probability that a type- θ sender has an accurate self-model. The key ordering is

$$1 - \delta > p_H > p_L > \bar{p} > \delta > 0.$$

Higher types are genuinely more self-accurate in the primitive environment. The lower benchmark $\bar{p}$ is the assessment induced by adverse repricing under high scrutiny.

This assumption should not be read as signal precision. Precision is a fidelity concept: a more precise audit reveals the true state more accurately. If high types are genuinely more accurate, a fidelity-improving audit tends to confirm them, not punish them. The model instead studies adverse social repricing. In a high-scrutiny state, an exposed self-presentation can be reinterpreted downward with a positive probability that is not driven by the sender's true merit. The economic loss is increasing in the amount of self-model reputation-at-risk carried by the true type.

Let $\alpha \in (0, 1]$ be the probability, conditional on high scrutiny and exposure, that the ambitious self-presentation is adversely repriced. This probability is taken to be independent of type in the baseline. It can be normalized into μ_H , but keeping it visible makes the interpretation clear: high scrutiny is not a deterministic truth audit. It is a state in which exposed self-presentations face a type-independent risk of adverse reinterpretation.

The epistemic exposure index is

$$\Phi_\theta = \alpha D_{\text{KL}}(\text{Bern}(p_\theta) \parallel \text{Bern}(\bar{p})).$$

The direction of the divergence reflects the economic object. The loss is not the ordinary probability of failure. It is the collapse from a carried self-accuracy level to a lower adverse benchmark. Judgment intensity prices this exposure:

$$C_E(\theta, j) = \mu_j \Phi_\theta.$$

Thus Φ_θ is neither history-induced reputation capital nor signal precision. It is the true-type-indexed loss from adverse repricing of self-model exposure. The receiver's later off-path attribution may affect the reputational return to a deviation, but it does not redefine Φ_θ .

Because the self-model prior enters this loss mechanically, higher prior confidence on accuracy creates larger Bayesian surprise if challenged later. In that sense, reputation-at-risk increases with prior. High prior can therefore generate fragility rather than extra safety.

3.3 Bayesian-surprise complementarity

Lemma 2 (Bayesian-surprise foundation for scrutiny-type complementarity). *Suppose*

$$1 - \delta > p_H > p_L > \bar{p} > \delta > 0.$$

Define

$$\Phi_\theta = \alpha D_{\text{KL}}(\text{Bern}(p_\theta) \parallel \text{Bern}(\bar{p}))$$

with $\alpha > 0$, and let

$$C_E(\theta, j) = \mu_j \Phi_\theta.$$

Then the epistemic cost has increasing differences in type and judgment intensity. For every $\mu' > \mu$,

$$C_E(\theta_H, \mu') - C_E(\theta_H, \mu) > C_E(\theta_L, \mu') - C_E(\theta_L, \mu).$$

Equivalently, the marginal cost of higher judgment intensity is larger for the high-self-accuracy type.

Proof. For Bernoulli beliefs,

$$D_{\text{KL}}(\text{Bern}(p) \parallel \text{Bern}(\bar{p})) = p \log \frac{p}{\bar{p}} + (1 - p) \log \frac{1 - p}{1 - \bar{p}}.$$

Differentiating with respect to p gives

$$\frac{\partial}{\partial p} D_{\text{KL}}(\text{Bern}(p) \parallel \text{Bern}(\bar{p})) = \log \frac{p(1 - \bar{p})}{\bar{p}(1 - p)}.$$

This derivative is positive whenever $p > \bar{p}$. Since $p_H > p_L > \bar{p}$, it follows that

$$\Phi_H > \Phi_L.$$

Therefore, for every $\mu' > \mu$,

$$[C_E(\theta_H, \mu') - C_E(\theta_H, \mu)] - [C_E(\theta_L, \mu') - C_E(\theta_L, \mu)] = (\mu' - \mu)(\Phi_H - \Phi_L) > 0.$$

Thus higher judgment intensity raises the epistemic cost of the high-self-accuracy type more than that of the low-self-accuracy type. $\square$

Remark 1 (Second-order self-model inference). *The key primitive behind $C_{\text{epistemic}}$ is not a new normative score; it is a Bayesian inference about self-judgment accuracy.*

A signal whose authenticity condition is neutrality to observation is strategically fragile: strategic deployment itself is an observation-seeking act and therefore updates beliefs toward "the signaler is managing being watched." That inference is stronger for higher-priority self-assessment priors.

Lemma 2 is the first reversal relative to standard signaling theory. In a Spence environment, high types are more willing to bear the ambitious signal because they face a lower ordinary cost of sending it. Here, high types also carry more self-model reputation-at-risk. When high scrutiny adversely reprices exposed self-presentation, the same type characteristic that supports ambitious action under low scrutiny becomes the source of larger exposure.

Remark 2 (Repricing, not precision). *The sign of Lemma 2 depends on the level channel. Replacing p_θ with signal precision would change the model. A higher-precision audit is a fidelity technology and tends to confirm an accurate high type. It therefore does not generate top-down withdrawal. The primitive needed here is a type-indexed self-accuracy level combined with an adverse repricing state, not a variance or precision parameter.*

3.4 From self-accuracy to crash risk

The preceding lemma converts primitive self-accuracy into exposure. A high type has more to gain from separation, but also more to lose from a future state that adversely reinterprets her ambitious signal as evidence about self-model risk.

Because Φ_θ is a true-type exposure cost, failure cost is not homogeneous across signal types: the ranking between types can and does reverse under repricing.

Let B_θ denote the current net separation rent from choosing s_A rather than s_P in the low-judgment state, before future high-scrutiny exposure is priced. Let

$$q = \Pr(j_{t+1} = H \mid j_t = L)$$

be the transition probability into high scrutiny. If choosing s_A creates the one-period shadow, the low-state payoff gain from choosing s_A can be written as

$$D_\theta^L(q) = B_\theta - \beta q \mu_H \Phi_\theta.$$

The associated threshold is

$$q_\theta^* = \frac{B_\theta}{\beta \mu_H \Phi_\theta},$$

whenever $B_\theta > 0$.

Under the maintained static-separation case $B_H > 0$ and $B_L < 0$, only the high type has a positive current rent from s_A . The operative threshold is therefore the high-type IC threshold

$$q_{IC} := q_H^* = \frac{B_H}{\beta \mu_H \Phi_H}.$$

For $q > q_{IC}$, the high type withdraws even while s_A is still read as a high-type signal. Whether pooling on s_P then survives off-path resurrection is the D1 question taken up in Section 4. The reversal relative to standard Spence is that the high type's constraint is the one that binds first: the same primitive that makes ambitious separation valuable also gives the high type larger exposure, $\Phi_H > \Phi_L$.

4 D1-Closed Belief Loops and Top-Down Pooling

The previous section established two ingredients. First, conditional on a fixed judgment-intensity state, the belief-dependent payoff is continuous in the relevant higher-order belief hierarchy. Hence the psychological-game object is well-defined within each regime slice. Second, the epistemic exposure term is pinned to the

sender's true type through primitive self-accuracy levels and adverse repricing. It is not a signal-precision parameter and not an equilibrium posterior.

This section closes the main argument. The local force is top-down: sufficiently likely future scrutiny makes high types less willing than low types to carry the ambitious signal, because the high type has more self-model reputation-at-risk under adverse repricing. The closed-loop force is inferential: once high types rationally withdraw, a D1-style off-path refinement no longer interprets an ambitious deviation as evidence of the high type. The result is a simultaneous consistency statement. For $q > q^*$, the only D1-closed pure assessment is safe pooling.

4.1 Anticipatory Incentive Failure

Let the current judgment-intensity state be low, denoted L , and let

$$q = \Pr(j_{t+1} = H \mid j_t = L)$$

be the transition probability into the high-scrutiny state. The sender chooses between the ambitious signal s_A and the pooling-safe signal s_P . Sending s_A creates an exposure state that can be repriced if the next-period state is H .

For type $\theta \in \{\theta_L, \theta_H\}$, let B_θ denote the current net separation surplus from sending s_A rather than s_P , before future high-scrutiny exposure is priced. This term includes the current reputational benefit of being interpreted as high type and subtracts current intrinsic and low-scrutiny epistemic costs. Let $\Phi_\theta > 0$ denote the true-type-indexed adverse-repricing exposure cost derived in Section 3. In the one-period exposure specification, the low-state payoff gain from sending s_A is

$$D_\theta(q) = B_\theta - \beta q \mu_H \Phi_\theta,$$

where $\beta \in (0, 1)$ is the discount factor and μ_H is the high-scrutiny intensity.

The high type's separating incentive constraint is

$$D_H(q) \geq 0.$$

Thus define

$$q_{IC} = \frac{B_H}{\beta \mu_H \Phi_H},$$

assuming $B_H > 0$. For $q > q_{IC}$, the high type strictly prefers the pooling-safe signal s_P to remaining on the ambitious signal s_A , even if s_A is still interpreted as high type.

Lemma 3 (Anticipatory failure of separation). *Suppose the current state is L . If*

$$q > q_{IC},$$

then the separating profile

$$\theta_H \mapsto s_A, \quad \theta_L \mapsto s_P$$

is not a psychological equilibrium.

Proof. Under the separating profile, s_A is interpreted as the high-type signal. Hence the high type receives the current separation surplus B_H from s_A , but also internalizes the future exposure cost $\beta q \mu_H \Phi_H$. Its payoff gain from choosing s_A rather than s_P is therefore

$$D_H(q) = B_H - \beta q \mu_H \Phi_H.$$

If $q > q_{IC}$, then $D_H(q) < 0$. Thus the high type strictly prefers s_P to s_A . The proposed separating profile violates the high type's incentive constraint and therefore cannot be an equilibrium. $\square$

The point is not that the low type successfully mimics the high type. The separating profile collapses before that question arises. The high type exits because the expected future cost of adverse repricing exceeds the current benefit of being recognized.

4.2 The D1 Flip

It remains to check whether pooling on s_P can be sustained. The possible obstruction is off-path resurrection: if all types pool on s_P , an off-path deviation to s_A might again be interpreted as evidence of high type, restoring the value of the ambitious signal.

This is where the refinement matters. Consider the pooling candidate in which both types choose s_P . The receiver observes an off-path signal s_A . Let a denote a receiver response, and let $R(a)$ be the reputational return from that response. For simplicity, assume $R(a)$ is type-independent and ordered by receiver favorability.

Let the cost of deviating to the ambitious signal for type θ be

$$K_\theta(q) = \bar{k}_\theta + \beta q \mu_H \Phi_\theta,$$

where $\bar{k}_\theta$ is the current cost of sending s_A in the pooling environment and $\beta q \mu_H \Phi_\theta$ is the anticipatory high-scrutiny exposure cost. The exposure term Φ_θ is the deviator's true-type cost. D1 determines how the receiver interprets the off-path signal and therefore affects $R(a)$. It does not relabel the deviator's exposure technology.

Given receiver response a , type θ 's payoff gain from deviating to s_A is

$$G_\theta(a; q) = R(a) - K_\theta(q).$$

Define the profitable-deviation response set

$$\mathcal{D}_\theta(q) = \{a : G_\theta(a; q) > 0\}.$$

In the standard Spence ranking, the high type has a lower signaling cost, so

$$\mathcal{D}_L(q) \subset \mathcal{D}_H(q).$$

D1 then assigns the off-path ambitious signal to the high type. The present model reverses this inclusion when future scrutiny is sufficiently likely. By Lemma 2, high types have larger future exposure:

$$\Phi_H > \Phi_L.$$

Hence the high type's cost increases faster in q . The ranking flips when

$$K_H(q) > K_L(q),$$

or equivalently,

$$\bar{k}_H + \beta q \mu_H \Phi_H > \bar{k}_L + \beta q \mu_H \Phi_L.$$

Define

$$q_{D1} = \frac{\bar{k}_L - \bar{k}_H}{\beta \mu_H (\Phi_H - \Phi_L)},$$

whenever $\bar{k}_L > \bar{k}_H$. Then for $q > q_{D1}$,

$$K_H(q) > K_L(q).$$

Lemma 4 (D1 flip). *Suppose $\Phi_H > \Phi_L$, $q > q_{D1}$, and the receiver-response space is rich enough that there exists a response a with*

$$K_L(q) < R(a) < K_H(q).$$

Then

$$\mathcal{D}_H(q) \subsetneq \mathcal{D}_L(q).$$

Therefore, under the D1 criterion, an off-path ambitious signal s_A cannot be assigned to the high type. It is assigned to the low type.

Proof. For any receiver response a , type θ 's gain from deviating to s_A is

$$G_\theta(a; q) = R(a) - K_\theta(q).$$

If $q > q_{D1}$, then $K_H(q) > K_L(q)$. Hence for every receiver response a ,

$$G_H(a; q) < G_L(a; q).$$

Therefore, whenever the deviation is profitable for the high type, it is also profitable for the low type:

$$G_H(a; q) > 0 \quad \Rightarrow \quad G_L(a; q) > 0.$$

Thus

$$\mathcal{D}_H(q) \subseteq \mathcal{D}_L(q).$$

The strict-response richness condition gives some a satisfying $K_L(q) < R(a) < K_H(q)$. For this response, $G_L(a; q) > 0$ while $G_H(a; q) < 0$, so $a \in \mathcal{D}_L(q) \setminus \mathcal{D}_H(q)$. Hence

$$\mathcal{D}_H(q) \subsetneq \mathcal{D}_L(q).$$

Under D1, if one type can profitably deviate under a strictly wider set of receiver responses, the deviation is attributed to that type rather than to the type whose profitable-deviation set is smaller. Hence the off-path signal s_A is attributed to θ_L , not to θ_H . $\square$

The ambitious signal therefore changes its meaning. In the standard model, an off-path ambitious signal says: this must be a strong type, because only strong types can afford it. Here, when future scrutiny is sufficiently likely, the same signal says: this is unlikely to be the high type, because high types have the most to lose from carrying exposed ambition into future scrutiny.

4.3 Belief Hierarchies and Closed Assessments

The fixed point is taken over a psychological assessment, not over a strategy profile alone. At a low-scrutiny history h_t , let

$$\sigma_\theta(h_t) \in \{s_A, s_P\}$$

denote the type- θ sender's signal choice. Let

$$\rho_A(h_t) := \Pr(\theta_H \mid s_A, h_t)$$

denote the receiver's first-order posterior after observing the ambitious signal. If s_A is on path, ρ_A is pinned down by Bayes' rule. If s_A is off path, ρ_A is selected by the refinement.

The sender's payoff depends on what she expects the receiver to believe after s_A . Write

$$\hat{\rho}_{\theta,A}(h_t)$$

for the type- θ sender's second-order belief about the receiver's posterior following s_A . A psychological assessment is a triple

$$\mathcal{A} = (\sigma, \rho_A, \hat{\rho}_A),$$

where $\hat{\rho}_A = (\hat{\rho}_{H,A}, \hat{\rho}_{L,A})$. Belief consistency requires

$$\hat{\rho}_{\theta,A} = \rho_A \quad \text{for each } \theta \in \{\theta_H, \theta_L\}.$$

Thus the sender's second-order belief about the receiver's first-order posterior must agree with the posterior induced by the assessment itself.

Crucially, the fixed point is evaluated exactly at this second-order hierarchy layer. To formally embed this within the framework of psychological games [1], the sender's expected utility must be continuous with respect to the relevant belief hierarchy. Here, the net benefit $B_\theta(\hat{\rho}_{\theta,A})$ depends continuously on the expected first-order posterior. While the D1 refinement injects a discontinuous jump into the receiver's belief map $\rho_A(\sigma)$ when the signal becomes off path, the fixed point is resolved on the composition of these expectations, so the belief loop closes at $\hat{\rho}_{\theta,A} = \rho_A$.

This is the level at which the fixed point is closed. The exposure term Φ_θ remains true-type indexed. It is not changed by the receiver's posterior. The receiver's posterior changes the current reputational benefit of the signal; it does not relabel the deviator's adverse-repricing exposure.

Let

$$B_\theta(\rho)$$

denote the type- θ current net benefit from choosing s_A rather than s_P when the ambitious signal is interpreted with posterior ρ . This term includes the current reputational return from the receiver's interpretation, net of current signal cost and low-scrutiny epistemic cost. It excludes future high-scrutiny exposure. The fixed B_H used above is the full high-type-attribution value:

$$B_H = B_H(1).$$

Assume

$$B_H(\rho) \text{ is weakly increasing in } \rho, \quad B_H(1) > 0, \quad B_H(0) \leq 0.$$

The first inequality says that a higher high-type posterior makes the ambitious signal more valuable. The last inequality says that an ambitious signal interpreted as low-type evidence carries no high-type premium. For the low type, assume

$$B_L(0) < 0.$$

Thus if s_A is interpreted as low-type evidence, the low type does not receive enough current benefit to justify choosing it.

The one-period exposure term $\beta q \mu_H$ can be written in the more general stock form used later. Let $e_t \geq 0$ denote exposed ambition at the beginning of period t . Sending s_A adds one unit of new exposed ambition; choosing s_P adds no new unit. Existing exposure depreciates at rate $\eta \in (0, 1]$:

$$e_{t+1} = (1 - \eta)e_t + \mathbf{1}\{s_t = s_A\}.$$

Let

$$\Gamma(q, \eta) > 0$$

denote the discounted marginal continuation price of adding one unit of exposed ambition in the current low-scrutiny state. The one-period reduced form is the special case

$$\Gamma(q, \eta) = \beta q \mu_H.$$

In the stock version,

$$\Gamma(q, \eta) = \mathbb{E}_q \left[\sum_{\tau \geq 1} \beta^\tau (1 - \eta)^{\tau-1} \mu_{j_t + \tau} \middle| j_t = L \right].$$

The maintained monotonicity condition is

$$\Gamma(q, \eta) \text{ is continuous and strictly increasing in } q.$$

Given posterior ρ , the high type's payoff gain from choosing s_A rather than s_P is

$$\Delta_H(\rho, q) = B_H(\rho) - \Gamma(q, \eta)\Phi_H.$$

Define the high-type withdrawal threshold by

$$q_{IC} = \inf\{q : B_H(1) - \Gamma(q, \eta)\Phi_H < 0\}.$$

In the one-period reduced form, this collapses to

$$q_{IC} = \frac{B_H(1)}{\beta\mu_H\Phi_H}.$$

Lemma 5 (Monotone stopping). *Suppose $B_H(\rho)$ is weakly increasing in ρ , $\Phi_H > 0$, and $\Gamma(q, \eta)$ is continuous and strictly increasing in q . Then the high type's stay advantage $\Delta_H(\rho, q)$ is strictly decreasing in q . In particular, if $q > q_{IC}$, then*

$$\Delta_H(\rho, q) < 0 \quad \text{for every } \rho \in [0, 1].$$

Thus the high type's withdrawal region is an upper interval in q .

Proof. For fixed ρ ,

$$\Delta_H(\rho, q) = B_H(\rho) - \Gamma(q, \eta)\Phi_H.$$

Since $\Phi_H > 0$ and $\Gamma(q, \eta)$ is strictly increasing in q , $\Delta_H(\rho, q)$ is strictly decreasing in q . Since $B_H(\rho) \leq B_H(1)$ for every $\rho \in [0, 1]$, the inequality

$$B_H(1) - \Gamma(q, \eta)\Phi_H < 0$$

implies

$$B_H(\rho) - \Gamma(q, \eta)\Phi_H < 0$$

for every $\rho \in [0, 1]$. Hence, once $q > q_{IC}$, no receiver posterior can make the high type willing to stay on s_A . $\square$

4.4 The Belief-Loop Operator

The off-path D1 comparison in Section 4.2 has the same form under the general exposure price:

$$K_\theta(q) = \bar{k}_\theta + \Gamma(q, \eta)\Phi_\theta.$$

Define

$$q_{D1} = \inf\{q : K_H(q) > K_L(q)\}.$$

For $q > q_{D1}$, the D1-selected off-path posterior after s_A is

$$\rho_A^{D1} = 0$$

under the strict-response richness condition of Lemma 4. The proof is the same as in Lemma 4, replacing $\beta q \mu_H$ by $\Gamma(q, \eta)$: once $K_H(q) > K_L(q)$, $G_H(a; q) < G_L(a; q)$ for every receiver response, and richness makes the inclusion strict. This D1 conclusion is an off-path counterfactual rule. It is obtained by comparing the response sets $\mathcal{D}_H(q)$ and $\mathcal{D}_L(q)$; it does not presuppose that either type actually deviates.

Let

$$z \in \{0, 1\}$$

denote whether the high type is expected to occupy the ambitious signal:

$$z = 1 \quad \text{means } \theta_H \text{ chooses } s_A, \quad z = 0 \quad \text{means } \theta_H \text{ withdraws to } s_P.$$

For $q > q_{D1}$, the receiver's belief about s_A is

$$\rho_A(z) = \begin{cases} 1, & \text{if } z = 1 \text{ and } s_A \text{ is the separating high-type signal,} \\ 0, & \text{if } z = 0 \text{ and } s_A \text{ is off path, selected by D1.} \end{cases}$$

The first line is Bayes' rule on the separating path. The second line is the D1 off-path counterfactual map.

Given this belief map, the high type's participation response is

$$T_q(z) = \mathbf{1} \{B_H(\rho_A(z)) - \Gamma(q, \eta)\Phi_H \geq 0\}.$$

A high-type belief-loop fixed point is a value $z \in \{0, 1\}$ such that

$$z = T_q(z).$$

This is the self-referential equation. High-type participation determines whether s_A is on path. The path status of s_A determines the receiver's posterior. The receiver's posterior determines the current rent of s_A . That rent determines the sender's second-order belief and high-type participation.

Define

$$q^* = \max\{q_{IC}, q_{D1}\}.$$

The closed-loop candidate that will survive is the safe-pooling configuration

$$\mathcal{C}_0 = (\sigma_H = s_P, \sigma_L = s_P; s_A \text{ off path; } \rho_A^{D1} = 0, \hat{\rho}_{H,A} = \hat{\rho}_{L,A} = 0).$$

It closes the loop because both types choose s_P , D1 evaluates the off-path s_A , D1 selects $\rho_A^{D1} = 0$, and sender second-order beliefs agree with that posterior.

4.5 Top-Down Pooling as the Unique D1-Closed Fixed Point

Theorem 1 (Top-down pooling as the unique D1-closed fixed point). *Suppose:*

1. $B_H(\rho)$ is weakly increasing in ρ , with $B_H(1) > 0$ and $B_H(0) \leq 0$;
2. $B_L(0) < 0$;
3. $\Phi_H > \Phi_L$, as derived from primitive self-accuracy levels and Bayesian-surprise repricing in Section 3;
4. $\Gamma(q, \eta)$ is continuous and strictly increasing in q , with $\eta \in (0, 1]$;
5. $q > q^* = \max\{q_{IC}, q_{D1}\}$;
6. off-path beliefs satisfy D1, with the strict-response richness condition of Lemma 4;

7. the analysis is restricted to the binary-signal pure-strategy class $\{s_A, s_P\}$.

Then no D1-consistent psychological assessment in this class has stable top-end occupancy of s_A . The unique D1-consistent pure closed-loop assessment is $\mathcal{C}_0$:

$$\theta_H \mapsto s_P, \quad \theta_L \mapsto s_P.$$

Equivalently, the unique high-type occupancy fixed point is

$$z^* = 0.$$

Proof. We evaluate all pure closed-loop configurations.

First, no configuration with top-end occupancy of s_A is feasible. If only the high type occupies s_A , then s_A is the separating high-type signal, so Bayes' rule gives

$$\rho_A(1) = 1.$$

The high type's payoff gain from choosing s_A rather than s_P is

$$\Delta_H(1, q) = B_H(1) - \Gamma(q, \eta)\Phi_H.$$

Since $q > q_{IC}$, this expression is strictly negative. Hence $T_q(1) = 0$, so $z = 1$ is not a fixed point.

If both types choose s_A , then s_A is pooling rather than separating. Let $\pi_0 \in (0, 1)$ be the prior probability of θ_H . Bayes' rule gives $\rho_A = \pi_0$. Since $B_H(\rho)$ is weakly increasing and $\pi_0 < 1$,

$$B_H(\pi_0) \leq B_H(1).$$

Because $q > q_{IC}$,

$$B_H(1) - \Gamma(q, \eta)\Phi_H < 0.$$

Therefore

$$B_H(\pi_0) - \Gamma(q, \eta)\Phi_H < 0.$$

The high type would prefer s_P , so pooling on s_A is not a fixed point either. Thus every configuration with top-end occupancy of s_A is infeasible.

Second, no low-only occupancy configuration is feasible. If only the low type chooses s_A , then s_A is on path as low-type evidence, so $\rho_A = 0$. The low type's gain is

$$B_L(0) - \Gamma(q, \eta)\Phi_L < 0,$$

because $B_L(0) < 0$ and $\Gamma(q, \eta)\Phi_L > 0$. Hence the low type does not wish to occupy s_A under low-type attribution.

Third, evaluate $\mathcal{C}_0$. Under $\mathcal{C}_0$, s_A is off path. Since $q > q_{D1}$, Lemma 4 and its general-price restatement imply

$$\mathcal{D}_H(q) \subsetneq \mathcal{D}_L(q),$$

so the off-path ambitious signal is assigned to the low type and

$$\rho_A^{D1} = 0.$$

Belief consistency then sets

$$\hat{\rho}_{H,A} = \hat{\rho}_{L,A} = \rho_A^{D1} = 0.$$

The high-type deviation margin is

$$\Delta_H^{D1}(q) = B_H(0) - \Gamma(q, \eta)\Phi_H < 0,$$

and the low-type deviation margin is

$$\Delta_L^{D1}(q) = B_L(0) - \Gamma(q, \eta)\Phi_L < 0.$$

Thus neither type has a profitable deviation to s_A . The safe-pooling configuration is internally consistent. Since all other pure configurations have been ruled out, $\mathcal{C}_0$ is the unique D1-consistent psychological assessment in the binary pure-strategy class. $\square$

The theorem should be read as a state claim:

$$\mathcal{C}_0 \iff [s_A \text{ off path}] \wedge [\rho_A^{D1} = 0] \wedge [\hat{\rho}_{H,A} = \hat{\rho}_{L,A} = 0] \wedge [\Delta_H^{D1}(q) < 0] \wedge [\Delta_L^{D1}(q) < 0].$$

The D1 component is not caused by an observed low-type deviation. It is the refinement's counterfactual response to an off-path signal, determined by $\mathcal{D}_H(q) \subsetneq \mathcal{D}_L(q)$. The participation inequalities then show that the counterfactual off-path interpretation is consistent with nobody actually sending s_A .

This separation removes the circular timing story. Reverse selection is the cost-ranking fact $\Phi_H > \Phi_L$. Individually rational withdrawal is the high-type inequality $\Delta_H(1, q) < 0$ once $q > q_{IC}$. Annihilation is the fixed-point conclusion that no D1-consistent pure assessment contains stable top-end occupancy of s_A once $q > q^*$.

In the coordination and self-fulfilling-belief literature, forward-induction refinement characteristically selects *toward* efficiency: it eliminates the off-path beliefs that sustain coordination failure, uniquely selecting the Pareto-dominant equilibrium [e.g., the refinement resolutions of 2]. This standard selection presumes that the party able to orchestrate a deviation is the one seeking the efficient outcome, so forward induction attributes the deviation's intent accordingly. We exhibit the opposite operation. When malicious scrutiny reverses the cost ranking so that the wider profitable-deviation set belongs to the *low* type, the same D1 logic confirms rather than dissolves the absence of stable top-end occupancy: the refinement selects safe pooling. This resulting top-down pooling is an *informational* disaster, destroying the market's ability to screen types. Yet it is also privately rational risk avoidance for high types facing excessive epistemic exposure. The refinement does not rescue the informational efficiency of the separating equilibrium; instead, it enforces the nonexistence of an occupied ambitious-top fixed point.

The theorem is deliberately pure-strategy. It does not use an interior mixing probability for the high type. In a binary model with no low-type background arrival at s_A , any strictly positive high-type mass on s_A would keep the posterior after s_A degenerate in the high type. Interior mixing requires an additional object, such as low-type mimetic entry, noisy audience inference, trembles, or a continuous type distribution. Those objects reappear as the polarization and population-dynamics implications below. The present section proves only the sharper fixed-point claim: above q^* , the only D1-consistent pure belief loop is $\mathcal{C}_0$, so the ambitious signal has no stable top-end occupancy fixed point.

5 Observable Implications: Polarization of Ambitious Signals

Section 4 closed the belief loop that makes safe pooling the unique survivor in the binary core. This section asks what that collapse looks like from the outside. The binary model delivers a sharp equilibrium statement: once the transition probability into high scrutiny exceeds the critical threshold, the high type exits the ambitious signal and pooling on the safe signal is selected under D1-consistent beliefs. This section records the observable implication in a richer type space. Near the threshold, the quality of the ambitious signal need not decline smoothly. The more distinctive prediction is support polarization: ordinary winners withdraw, while low-calibration agents and extreme high types may remain.

5.1 Continuous types

Let the type space be an interval

$$\Theta = [\underline{\theta}, \bar{\theta}].$$

The sender chooses between s_A and s_P . The current low-scrutiny net separation rent from s_A is $B(\theta)$. The true-type-indexed adverse-repricing exposure is $\Phi(\theta)$, derived from primitive self-accuracy levels

$$p(\theta) = \Pr(m = 1 \mid \theta), \quad p'(\theta) > 0,$$

through

$$\Phi(\theta) = \alpha D_{\text{KL}}(\text{Bern}(p(\theta)) \parallel \text{Bern}(\bar{p})).$$

The low-state payoff gain from the ambitious signal is

$$D(\theta, q) = B(\theta) - \beta q \mu_H \Phi(\theta).$$

Thus type θ chooses s_A exactly when

$$q < q^*(\theta), \quad q^*(\theta) = \frac{B(\theta)}{\beta \mu_H \Phi(\theta)}.$$

The binary theorem corresponds to the case in which the top type's threshold is crossed first. In the continuous model, the entire shape of $q^*(\theta)$ determines which types remain on the ambitious signal.

5.2 Hollowing out

A simple mean-reversion story would predict that, as scrutiny risk rises, the ambitious signal gradually becomes lower quality. That is not the generic implication of top-down pooling. The model removes types according to the exposure-to-rent ratio

$$\frac{\Phi(\theta)}{B(\theta)},$$

not according to type itself.

The resulting profile is density-geometric: the mode of the conditional distribution can lie strictly below the turning threshold, while the maximally safe position is depopulated. The most crowded point need not be the most stable one.

Suppose $q^*(\theta)$ is nonmonotone: it is low on an intermediate-high region of the type space but rises again in the extreme upper tail. Economically, these are environments in which ordinary winners carry substantial reputation-at-risk but do not have enough surplus to dominate future adverse repricing, while extreme types have sufficiently large current surplus or sufficiently robust self-models to remain exposed.

For a given q , the set of types choosing the ambitious signal is

$$A(q) = \{\theta \in \Theta : q < q^*(\theta)\}.$$

If $q^*(\theta)$ has an interior trough, then for transition probabilities in the corresponding range,

$$A(q) = [\underline{\theta}, \theta_1(q)] \cup [\theta_2(q), \bar{\theta}]$$

for some $\theta_1(q) < \theta_2(q)$. The ambitious signal is then generated by two separated groups: low or poorly calibrated types whose exposure-to-rent ratio is low because they have little to lose, and extreme high types whose rents remain large enough to bear exposure.

Proposition 1 (Polarized degradation). *Suppose $q^*(\theta)$ is continuous and there exist $\theta_1 < \theta_2$ such that*

$$q^*(\theta) > q \quad \text{for } \theta \in [\underline{\theta}, \theta_1] \cup [\theta_2, \bar{\theta}]$$

and

$$q^*(\theta) < q \quad \text{for } \theta \in (\theta_1, \theta_2).$$

Then the type distribution conditional on the ambitious signal has support on two separated intervals. In particular, the ambitious signal is not a monotone high-type signal in this region.

Proof. The sender chooses s_A if and only if $q < q^*(\theta)$. The stated inequalities imply that this condition holds exactly on $[\underline{\theta}, \theta_1] \cup [\theta_2, \bar{\theta}]$ and fails on (θ_1, θ_2) . Hence the support of θ conditional on observing s_A is the union of two separated intervals. This support is not an upper set in Θ . Therefore s_A is not a monotone high-type signal. $\square$

The proposition is a support result, not a variance theorem. It says that the ambitious signal can cease to be an upper set in type space. The conditional mean of type among ambitious senders may fall, but the model does not imply an unconditional sign for

$$\text{Var}(\theta \mid s_A).$$

A variance increase is an empirical implication only under regular density and mass-balance conditions. For example, suppose the underlying type density is continuous, both surviving tail intervals retain positive conditional mass, the removed middle interval has its conditional mean between the two surviving tail means, and the separation between the two surviving tail means dominates the within-tail dispersion introduced by truncation. Under such conditions, hollowing out the middle raises conditional dispersion relative to the pre-threshold ambitious set. Without those conditions, the sign of the variance comparison is an empirical question.

The robust prediction is therefore polarization of support: the surviving ambitious signal can be generated by extreme competence or by poor anticipatory discipline. This is why mean reversion is too weak a description. The ambitious signal becomes nonmonotone before any unconditional variance claim is available.

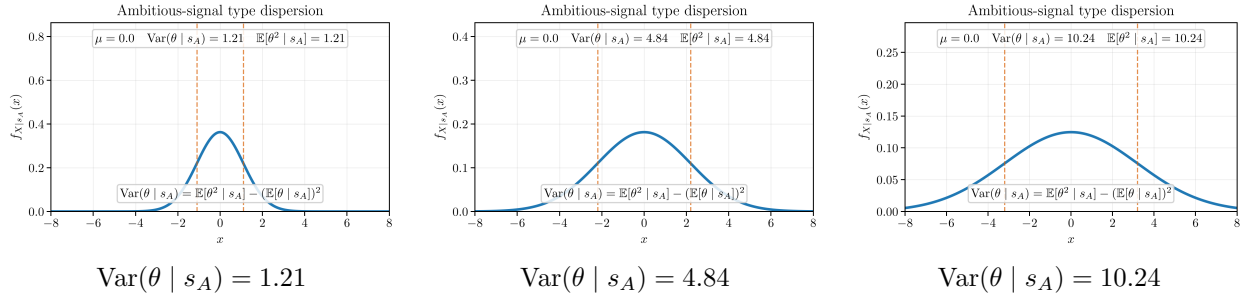


Figure 1: Illustrative simulations under regular mass-balance conditions: conditional dispersion can rise for the ambitious signal.

5.3 Relation to the D1 flip

The D1 mechanism is the one proved in Section 4, including its contrast with standard forward-induction selection after Theorem 1. The present section uses that result only as an observable implication: once the high type has the smaller profitable-deviation response set, the ambitious signal can become nonmonotone rather than simply “high type.”

5.4 Empirical language

The model predicts a specific kind of degradation. It is not merely that ambitious signals become noisier. It is that the middle of the credible ambitious class exits first. Agents with too little reputation-at-risk may continue because adverse repricing cannot take much from them. Agents with overwhelming surplus may continue because the ambitious signal remains worthwhile. Agents in between withdraw because they have enough to lose and not enough surplus to dominate the future exposure cost.

Hence the signal quality does not just mean-revert; it can deplete from the center and leave a polarized residual set at the margin.

In the standard signaling model, the high signal is protected because bad types cannot climb up. In this model, the high signal is hollowed out because good types climb down. The remaining signal can look extreme: true outliers and badly calibrated imitators share the same visible action. That polarization is the observable signature of top-down pooling.

This polarized, hollowed signature is the static cross-sectional shadow of top-down pooling: a snapshot of who still sends the ambitious signal once withdrawal has run its course.

6 Dynamic Forward Invariance

Section 4 establishes that in the binary class, the one-period object selects pooling for $q > q^*$. This section adds the missing microfounded dynamic state and asks whether endogenous accumulation can reverse that outcome.

6.1 Trust-capital law of motion

Let (C_t, E_t) be a two-dimensional endogenous state observed before j_t is realized.

- C_t is semantic trust capital created by repeated pooling-safe behavior.
- E_t is engagement reinforcement (habituation) accumulated from sustained non-ambitious actions.

The state evolution is

$$\begin{aligned} C_{t+1} &= (1 - \rho_C)C_t + \eta_C \mathbf{1}\{s_t = s_P\} - \psi_C \mathbf{1}\{s_t = s_A\} - \xi_t^C, \\ E_{t+1} &= (1 - \rho_E)E_t + \eta_E \mathbf{1}\{s_t = s_P\} + \eta_A \mathbf{1}\{m_t = 1\} - \xi_t^E, \end{aligned}$$

where $m_t \in \{0, 1\}$ is a cheap trust-restoring message available to both types and ξ_t^C, ξ_t^E are bounded asynchronous shocks. This keeps the mechanism endogenous while preserving one-period exposure at the signaling margin.

Define a temporary semantic penalty term

$$\Sigma_t = \omega_C C_t + \omega_E E_t \geq 0.$$

Then, in low-scrutiny state L , the low-state advantage of sending ambitious is

$$\Delta_\theta^{\text{dyn}}(q_t, \rho_t) = B_\theta(\rho_t) - \beta q_t \mu_H \Phi_\theta - \Sigma_t,$$

where ρ_t is the period- t version of the Section 4 posterior ρ_A and appears in B_θ . Sending s_P avoids the future repricing term and does not create today's semantic penalty.

6.2 Unpriced trust-restoring messages

Cheap anthropomorphic messages are unpriced: the sender can choose $m_t = 1$ with no direct signal-dependent cost and no type restriction. Because both types can choose m_t identically, the induced change in (C_t, E_t) is type-neutral. Therefore these messages are shadowable.

To see dynamic single crossing, let receiver response a induce continuation return $R(a)$ from choosing s_A . The high- and low-type off-path gains are

$$G_H(a; t) = R(a) - \bar{k}_H - \beta q_t \mu_H \Phi_H - \Sigma_t, \quad G_L(a; t) = R(a) - \bar{k}_L - \beta q_t \mu_H \Phi_L - \Sigma_t.$$

The term Σ_t cancels in the comparison, so adding or relaxing trust-capital does not alter the type comparison. If $q_t > q_{D1}$, then $G_H(a; t) < G_L(a; t)$ for every receiver response, so s_A is still D1-attributed to the low type.

Remark 3 (Invariance of the Belief Loop Fixed Point). *Embedding the dynamic penalty Σ_t into the belief-loop operator $T_q(z)$ from Section 4 does not alter the number of fixed points in the collapse region. Let the modified operator be $T_{q, \Sigma_t}(z) = \mathbf{1}\{B_H(\rho_A(z)) - \beta q_t \mu_H \Phi_H - \Sigma_t \geq 0\}$. Because $\Sigma_t \geq 0$, $T_{q, \Sigma_t}(0) = 0$ remains robustly true, as $B_H(0) \leq 0$ and the penalty only pushes the deviation payoff further negative. For the separating candidate ($z = 1$), we have $T_{q, \Sigma_t}(1) = 1 \iff \Sigma_t \leq B_H(1) - \beta q_t \mu_H \Phi_H$. In the collapse region ($q_t > q_{IC}$), the right-hand side is already strictly negative; thus, any $\Sigma_t \geq 0$ only deepens the impossibility of separation, ensuring $z^* = 0$ remains the unique fixed point. If the system is subcritical ($q_t < q_{IC}$), an extreme dynamic shock Σ_t could trigger a premature collapse by destroying the $z = 1$ fixed point, but dynamic penalties can never create a new fixed point nor resurrect separation.*

6.3 Dynamic Spence ratio and wedges

For any endogenous state (C_t, E_t) and posterior ρ_t , define the state-adjusted top-down ratio

$$\tilde{q}_\theta^{\text{IC}}(t) = \frac{B_\theta(\rho_t) + \Sigma_t}{\beta \mu_H \Phi_\theta}.$$

Endogenous trust-capital terms appear symmetrically in numerators and therefore do not change the fundamental type-ordering created by (Φ_H, Φ_L) and static primitives. They shift the threshold pair, but not the sign of the top-down reversal.

A true restoration requires an externally priced type-specific wedge, for example a direct penalty or premium that depends on sender type/claim context and is not a pure continuation by-product of C_t and E_t .

6.4 Forward invariance

Let

$$\mathcal{Q} := \{q \in [0, 1] : q \geq q^*\}, \quad q^* = \max\{q_{IC}, q_{D1}\}.$$

Assume that under pooling on s_P , the process satisfies:

1. $\Sigma_{t+1} - \Sigma_t \geq \underline{d} - \xi_t$ for some $\underline{d} > 0$ and bounded $\xi_t \in [0, \bar{\xi}]$;
2. $\Sigma_0 \geq 0$ and $\bar{\xi} < \underline{d}$;
3. $\beta q \mu_H \Phi_H - B_H(0) - \Sigma_0 > 0$ for all $q \in \mathcal{Q}$.

Theorem 2 (Pooling interval forward invariance). *Under these assumptions and strict D1 preferences at $q \in \mathcal{Q}$, if period t is in the pooling assessment of Section 4, then period $t + 1$ is also in that assessment with probability one. Equivalently, $\mathcal{Q}$ is almost surely forward-invariant.*

Proof. In the pooling assessment, $\rho_t = 0$ for any off-path ambitious deviation. The high-type deviation margin is

$$\Delta_H^{\text{dyn}}(q_t, 0) = B_H(0) - \beta q_t \mu_H \Phi_H - \Sigma_t.$$

By assumption 3 and $\Sigma_t \geq \Sigma_0$, $\Delta_H^{\text{dyn}}(q_t, 0)$ is negative for all $q_t \in \mathcal{Q}$. Assumption 1 says endogenous drift in Σ_t dominates bounded asynchronous noise, so this negativity cannot be overturned by one-period shocks. Hence high type never has a profitable ambiguous-ambitious deviation from s_P once in the pooling band, and D1 therefore continues to map s_A to low attribution. The same argument applies period-by-period by induction. $\square$

The conclusion is defensive. Dynamic trust-capital and habituation can be modeled explicitly and can affect local margins, but they do not create a new stopping phenomenon. They do not by themselves repair top-down collapse.

7 Continuous-Time Population Dynamics and Top-End Depletion

Section 4 gives the strategy-level statement: for $q > q^*$, no D1-consistent pure assessment has stable top-end occupancy of the ambitious signal. This section gives the distribution-level counterpart. It treats ambitious participation as a continuous population stock and asks what the cross section looks like once the same D1-closed payoff map is imposed.

The correspondence is:

$$\text{no occupied ambitious fixed point} \iff \text{demographic exit from the top-end ambitious pool.}$$

The continuous model is not a chronological avalanche story. It is an economic packaging of the same closed-loop state claim: D1 closure turns top-end ambitious participation into an exit process, and the stationary boundary profile records where the remaining ambitious density can be sustained.

7.1 Reaction-Diffusion Occupancy

Let $x \in [0, \infty)$ denote a continuous economic state. Higher x means higher visible status, stronger reputation-at-risk, or more self-accurate exposure to future adverse repricing. Let $n(x, t) \geq 0$ be the mass density of agents currently occupying the ambitious signal s_A at state x . The relevant object is the D1-closed aggregate ambitious pool, not separate type-by-type participation.

Local mobility in status space is summarized by a diffusion term. Local entry, retention, and exit from the ambitious pool are summarized by the D1-closed growth rate

$$V_q(x) = B^{D1}(x) - \Gamma(q, \eta)\Phi(x).$$

Here $B^{D1}(x)$ is the current net return to ambitious occupancy after the off-path D1 map has selected the low-attribution assessment, while $\Gamma(q, \eta)\Phi(x)$ is the continuation exposure price inherited from Section 4. The function $\Phi(x)$ is increasing enough near the top that

$$V_q(x) < 0$$

on the exposed upper region. In that region the demographic exit rate from the ambitious pool is

$$\kappa_q(x) := -V_q(x) > 0.$$

The resulting continuous-time occupancy equation is

$$\frac{\partial n}{\partial t} = \frac{\sigma^2}{2} \frac{\partial^2 n}{\partial x^2} + V_q(x)n. \quad (1)$$

The first term lets ambitious density diffuse across nearby economic states. The second term is birth-death accounting inside the ambitious pool: $V_q(x) > 0$ permits local replenishment, while $V_q(x) < 0$ kills ambitious occupancy at rate $\kappa_q(x)$. The D1 closure matters because it changes the top-end payoff from a reputational rent into a negative participation margin.

7.2 Long-Run Cross-Sectional Density

A stable long-run cross-sectional density has the form

$$n(x, t) = e^{\lambda t} \psi(x), \quad \psi(x) \geq 0,$$

with normalization supplied by the protected lower region. Substituting this form into equation (1) gives

$$\frac{\sigma^2}{2} \psi''(x) + V_q(x) \psi(x) = \lambda \psi(x). \quad (2)$$

The scalar λ is the net long-run growth rate of the ambitious stock under the D1-closed occupancy process. When the remaining density is normalized to unit mass, ψ is the stable long-run cross-sectional shape.

The boundary conditions have direct economic meanings. At the protected lower limit, the model either anchors base density or fixes a normalization:

$$\psi(0) = \psi_0 > 0 \quad \text{or} \quad \int_0^\infty \psi(x) dx = 1.$$

At the far top end, the non-explosive transversality condition rules out profiles that create infinite ambitious mass:

$$\limsup_{x \rightarrow \infty} \psi(x) < \infty, \quad \psi \in L_+^1(0, \infty).$$

If the upper tail has $V_q(x) \leq -\underline{\kappa} < 0$, this condition selects a decaying boundary profile. Population density cannot be negative, and an admissible long-run cross section cannot alternate signs. Oscillatory continuations are therefore not economic population densities.

7.3 Critical Boundary Profile

Let $\bar{x}$ be the critical top boundary where the D1-closed growth rate changes sign relative to the long-run growth rate. Near that boundary, use the turning-point approximation

$$V_q(x) \approx a(q) - b(\bar{x} - x), \quad b > 0.$$

With the inward coordinate

$$y = \bar{x} - x,$$

equation (2) becomes

$$\frac{\sigma^2}{2} \psi''(y) + (a(q) - by) \psi(y) = \lambda \psi(y). \quad (3)$$

After the rescaling

$$z = \left(\frac{2b}{\sigma^2} \right)^{1/3} \left(y - \frac{a(q) - \lambda}{b} \right),$$

the local equation takes the standard turning-point form

$$\frac{d^2\psi}{dz^2} - z\psi = 0.$$

Mathematically, this is the canonical Airy turning-point solution. Economically, only the nonnegative bounded branch is admissible: the growing branch violates non-explosive transversality, and the oscillatory continuation would imply negative population density. The relevant object is the nonnegative lobe on the protected side together with the decaying boundary-layer tail on the top side.

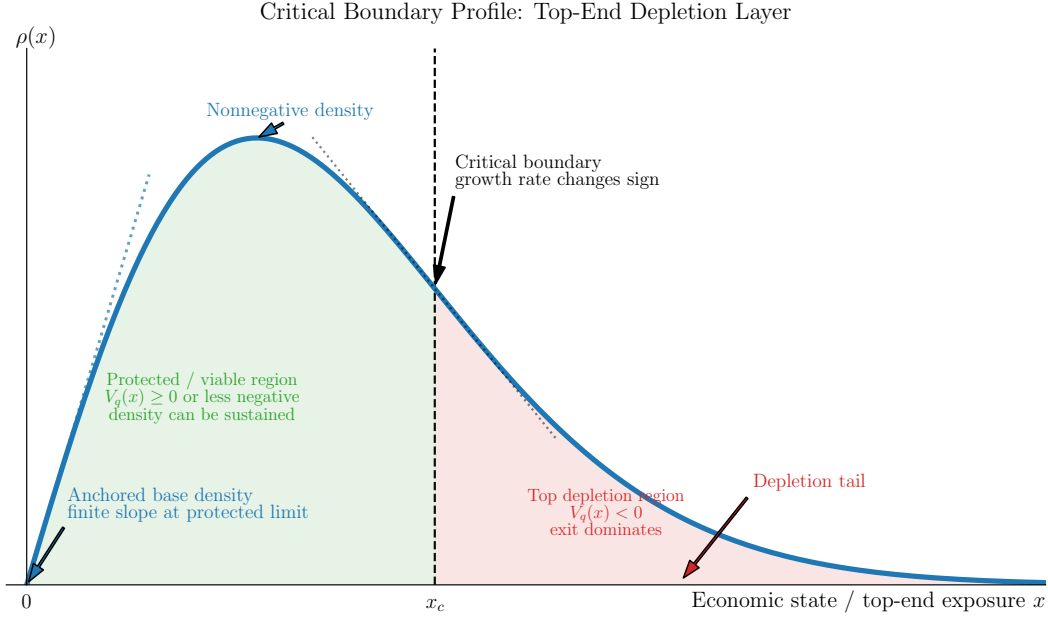


Figure 2: Critical boundary profile. Near the cutoff where the D1-closed growth rate changes sign, the stationary occupancy equation has the standard turning-point form. The decaying boundary-layer profile represents depletion of ambitious occupancy at the top end.

Figure 2 is a local critical snapshot, not a new global equilibrium theorem. The protected side represents states where ambitious occupancy can be sustained or replenished. The exposed top side represents states where $V_q(x) < 0$, so demographic exit dominates local diffusion and the density thins into a depletion tail.

7.4 Economic Implication

The continuous population formulation adds one distributional implication to the fixed-point theorem. D1 closure removes the high-type rent from ambitious occupancy; at sufficiently exposed top states, the remaining D1-closed payoff is negative. The reaction-diffusion process then has a top-end depletion layer rather than a refilled ambitious upper tail.

Thus the continuous model is the macroscopic shadow of the belief-loop fixed point. Strategy level: no D1-consistent fixed point has stable top-end occupancy of s_A . Population level: once $V_q(x) < 0$ near the critical top boundary, ambitious density is killed in that region and the admissible long-run profile decays. Continuous population diffusion does not refill top-end ambitious occupancy once the D1-closed exit rate dominates.

References

- [1] P. Battigalli and M. Dufwenberg. Dynamic signaling in psychological games. *Journal of Economic Theory*, 144(2):523–543, 2009.
- [2] D. W. Diamond and P. H. Dybvig. Bank runs, deposit insurance, and liquidity. *Journal of Political Economy*, 91(3):401–419, 1983.